LSTM Questions – Question Answer Bank

Question 1: Single-Cell Forward Pass

Given:

- $x_t = 0.5$
- $h_{t-1} = 0.2$
- $C_{t-1} = 0.4$

We calculate the forget gate, input gate, candidate cell state, new cell state, output gate, and finally the hidden state.

1. Forget Gate

$$f_t = \sigma(W_f x_t + U_f h_{t-1} + b_f)$$

$$= \sigma[0.7(0.5) + 0.3(0.2) + 0.1]$$

$$= \sigma(0.35 + 0.06 + 0.10)$$

$$= \sigma(0.51)$$

$$f_t = 0.6248$$

2. Input Gate

$$i_t = \sigma(W_i x_t + U_i h_{t-1} + b_i)$$

$$= \sigma[0.6(0.5) + 0.4(0.2) + 0]$$

$$= \sigma(0.30 + 0.08)$$

$$= \sigma(0.38)$$

$$i_t = 0.5939$$

3. Candidate Cell State

The candidate cell state uses **tanh** instead of sigmoid.

$$\tilde{C}_t = \tanh(W_c x_t + U_c h_{t-1} + b_c)$$

$$= \tanh[0.9(0.5) + 0.2(0.2)]$$

$$= \tanh(0.45 + 0.04)$$

$$= \tanh(0.49)$$

$$\tilde{C}_t = 0.4542$$

4. New Cell State

$$C_t = f_t C_{t-1} + i_t \tilde{C}_t$$

$$= (0.6248)(0.4) + (0.5939)(0.4542)$$

$$= 0.2499 + 0.2697$$

$$C_t = 0.5197$$

5. Output Gate

$$o_t = \sigma(W_o x_t + U_o h_{t-1} + b_o)$$

$$= \sigma[0.5(0.5) + 0.5(0.2) + 0.1]$$

$$= \sigma(0.25 + 0.10 + 0.10)$$

$$= \sigma(0.45)$$

$$o_t = 0.6106$$

6. Hidden State

$$h_t = o_t \tanh(C_t)$$

$$= 0.6106 \times \tanh(0.5197)$$

$$= 0.6106 \times 0.4775$$

ht = 0.2916

Final Answer:

Value	Answr
f_t	0.6248
i_t	0.5939
$\tilde{C}_t$	0.4542
C_t	0.5197
o_t	0.6106
h_t	0.2916

Question 2: LSTM Parameter Count

An LSTM layer has:

- Input size = **10**
- Hidden units = **32**

There are four components in an LSTM:

1. Forget gate
2. Input gate
3. Candidate cell state
4. Output gate

For each gate:

$$W = 32 \times 10 = 320$$
$$U = 32 \times 32 = 1024$$

(a) One Bias Vector Per Gate

Bias parameters:

$$b = 32$$

Therefore, parameters per gate:

$$320 + 1024 + 32 = 1376$$

For four gates:

$$4 \times 1376$$
5504

Answer: 5,504 trainable parameters.

(b) PyTorch Convention – Two Bias Vectors

Here, each gate has two bias vectors.

$$32 + 32 = 64$$

Parameters per gate:

$$320 + 1024 + 64 = 1408$$

For four gates:

$$4 \times 1408$$
5632

Answer: 5,632 trainable parameters.

Question 3: Vanishing Gradient Through Cell State

The gradient scaling is approximately:

$$f^k$$

where f is the forget-gate value and k is the number of time steps.
LSTM_Practice_Questions

(a) For $f = 0.9$

For 10 steps:

$$0.9^{10} = 0.3487$$

For 50 steps:

$$0.9^{50} = 0.0052$$

Answer:

0.3487 after 10 steps
0.0052 after 50 steps

(b) For 50 Steps

For $f = 0.5$:

$$0.5^{50} = 8.88 \times 10^{-16}$$

This value is almost zero.

For $f = 0.99$:

$$0.99^{50} = 0.6050$$

(c) Explanation

When the forget-gate value is small, the gradient decreases very quickly. For example, with $f = 0.5$, almost no gradient survives after 50 steps.

When $f = 0.99$, around **60.5%** of the gradient remains. Therefore, an LSTM can learn long-term dependencies by keeping its forget gate close to 1 when information needs to be remembered.

Compared with a vanilla RNN, the LSTM cell state provides a better path for information and gradients to travel across many time steps.

Question 4: Electricity Load Forecasting

The utility has:

- 3 years of data
- 365 days per year
- 24 hours per day
- Lookback = 168 hours
- Prediction = next 24 hours
- Features = 3

These are the conditions specified in the question. LSTM_Practice_Questions

(a) Total Hourly Records

$$3 \times 365 \times 24$$

26,280

Answer: 26,280 hourly records.

(b) Number of Windows

Each complete window requires:

$$168 + 24 = 192 hours$$

Therefore:

$$26,280 - 192 + 1$$

26,089

Answer: 26,089 windows.

(c) Tensor Shapes

For a batch size of 64:

Input:

64,168,3

where:

- 64 = batch size
- 168 = time steps
- 3 = features

Output:

64,24

because we predict 24 future load values.

(d) Training Windows

70% of records are used for training.

$$0.70 \times 26,280 = 18,396$$

Therefore:

$$18,396 - 192 + 1$$

18,205 training windows

Now divide by batch size:

$$18,205/64 = 284.45$$

So there are:

- 284 complete batches
- 1 partial batch

285 mini-batches per epoch

(e) LSTM Parameters

Input features:

$$m = 3$$

Hidden size:

$$n = 64$$

Formula:

$$4[(n \times m) + (n \times n) + n]$$

Substituting:

$$4[(64 \times 3) + (64 \times 64) + 64]$$

$$= 4[192 + 4096 + 64]$$

$$= 4(4352)$$

17,408

Answer: 17,408 trainable parameters.

Question 5: Predictive Maintenance

The vibration readings range from **20 to 120 mm/s**, and the LSTM predicts whether the pump will fail in the next cycle. LSTM_Practice_Questions

(a) Min-Max Normalisation

Formula:

$$x_{norm} = \frac{x - x_{min}}{x_{max} - x_{min}}$$

For $x = 95$:

$$= \frac{95 - 20}{120 - 20}$$

$$= \frac{75}{100}$$

0.75

(b) Model Prediction

Model probability:

$$0.83$$

Threshold:

$$0.5$$

Since:

$$0.83 > 0.5$$

the model predicts:

Failure

(c) Confusion Matrix

Total cycles = **1000**

Actual failures = **40**

Predicted failures = **45**

Correct failure predictions = **30**

Therefore:

$$TP = 30$$

$$FP = 45 - 30 = 15$$

$$FN = 40 - 30 = 10$$

$$TN = 1000 - 30 - 15 - 10$$

$$TN = 945$$

	Predicted Failure	**Predicted OK**
Actual Failure	TP = 30	FN = 10
Actual OK	FP = 15	TN = 945

Precision

$$Precision = \frac{TP}{TP+FP}$$

$$= \frac{30}{30+15}$$

$$= \frac{30}{45}$$

0.6667=66.67%

Recall

$$Recall = \frac{TP}{TP+FN}$$

$$= \frac{30}{30+10}$$

$$= \frac{30}{40}$$

0.75=75%

F1-Score

$$F1 = \frac{2PR}{P+R}$$

$$= \frac{2(0.6667)(0.75)}{0.6667+0.75}$$

0.7059=70.59%

Accuracy

$$Accuracy = \frac{TP+TN}{Total}$$

$$= \frac{30+945}{1000}$$

$$= \frac{975}{1000}$$

0.975=97.5%

(d) Which Metric Should Be Prioritised?

The maintenance team should prioritise **recall** because missing an actual pump failure can result in unexpected breakdowns, production loss, and costly repairs.

Accuracy can be misleading because only **40 out of 1,000 cycles (4%)** are failures. Even if a model predicted **"no failure" for every cycle**, it would still achieve:

9601000times100=96%

accuracy while detecting **zero actual failures**.

Therefore, **recall is more important**, while precision and F1-score should also be monitored.